

Requirements

1. Overview

This document lists all software dependencies required to run the Configuration Performance Learning tool. The tool is implemented entirely in Python and uses standard scientific computing libraries.

2. Python Version

Required: Python 3.9 or higher (Python 3.11 recommended)

Download from: <https://www.python.org/downloads/>

3. Required Python Libraries

Install all dependencies using the following command:

- `pip install scikit-learn pandas numpy scipy`

Library	Version	Purpose
scikit-learn	<code>>=1.3.0</code>	Linear Regression, Random Forest Regressor, Gaussian Process Regressor, Stand and Scaler, metrics
pandas	<code>>=2.0.0</code>	CSV loading, DataFrame operations, result aggregation
numpy	<code>>=1.24.0</code>	Numerical operations, array handling
scipy	<code>>=1.11.0</code>	Wilcoxon signed-rank statistical test

4. Operating System

The tool has been developed and tested on **Windows 11**. It is compatible with any OS that supports Python 3.9+, including macOS and Linux.

5. No Additional Installation Required

The tool does not require any compilation step, external binaries, or database setup. All datasets are provided as CSV files in the **datasets/** directory cloned from the course GitHub repository (<https://github.com/ideas-labo/ISE/tree/main/lab2>).

6. Version Verification

To verify installed versions run:

- `python -c "import sklearn, pandas, numpy, scipy; print(sklearn.__version__, pandas.__version__, numpy.__version__, scipy.__version__)"`